Prepared for submission to Riverty Group GmbH

One Student One Laptop (OSOL)

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Contents

1 What is One Student One Laptop(OSOL)? 4

2 Why OSOL? 4
- 2.1 Education would change one life 4
  - 2.1.1 What is Education and Why is it Important? 4
  - 2.1.2 How Education Can Empower Individuals and Communities 5
- 2.2 The Internet and Education 5
  - 2.2.1 Unprecedented Access to Information 6
  - 2.2.2 Enhanced Communication and Collaboration 6
  - 2.2.3 Expanded Access to Educational Resources 6
  - 2.2.4 Online Learning and Certification 6
  - 2.2.5 Flexible Learning Environments 6
  - 2.2.6 Beyond the Classroom 7
  - 2.2.7 Cost-Effective Learning Opportunities 7
- 2.3 The Impact of Artificial Intelligence on Education 7
  - 2.3.1 Personalized Learning Experiences 7
  - 2.3.2 24/7 AI-Powered Assistance 7
  - 2.3.3 Enhanced Learning and Reduced Errors 7
  - 2.3.4 Streamlining Tasks and Fostering Independent Learning 7

3 Related Resources 8
- 3.1 One Laptop Per Child 8
- 3.2 Labdoo Deutschland 8
- 3.3 World Computer Exchange 9
- 3.4 Computer Aid International 9
- 3.5 Computer for School in Kenya 10
- 3.6 Camara Education 11
- 3.7 Close the Gap 11
- 3.8 Open Educational Resources 12
- 3.9 eGranary Digital Library 13
- 3.10 UNESCO 14
  - 3.10.1 The Importance of Digital Innovation in Education 14
  - 3.10.2 The Role of Artificial Intelligence in Education 15

4 What OSOL will benfit to Riverty and Bertelsmann? 15

5 How OSOL? 17
- 5.1 One IT equipment lifecycle 17
  - 5.1.1 Equipment stages 17
  - 5.1.2 Equipment score 17
  - 5.1.3 Equipment annual check 18
- 5.2 OSOL 2D barcode on IT equipment 18
  - 5.2.1 Stage0 2D barcode data 18
  - 5.2.2 Stage1 2D barcode data 18

5.2.3 Stage2 2D barcode data 20
5.2.4 Stage3 2D barcode data 21
5.3 OSOL preinstalled education content 21
5.3.1 Online education content 21
5.3.2 Offline education content 21
5.4 OSOL Tracer - OSOL Data Management System 21
5.4.1 OSOL-T Architecture 21
5.4.2 OSOL-T Database 22
5.4.3 OSOL-T API 23
5.4.4 OSOL-T API Management 23
5.4.5 OSOL-T GUI 24
5.5 Riverty Computer Center(RCC) 25

6 When OSOL - Schedule 25
6.1 Step 1 25

7 OSOL Budget 25

A OSOL-T Database metadata 26
A.1 table_audit_columns 26
A.2 osol.person 26
A.3 conf.person_type 26
A.4 conf.language 27
A.5 osol.address 27
A.6 conf.country 28
A.7 osol.communication 28
A.8 osol.communication_mean 28
A.9 osol.equip 29
A.10 conf.equip_type 29
A.11 osol.monitor 30
A.12 osol.monitor_model 30
A.13 osol.keyboard 31
A.14 osol.keyboard_model 31
A.15 osol.mouse 32
A.16 osol.mouse_model 32
A.17 osol.laptop 32
A.18 osol.laptop_model 33
A.19 osol.cpu_model 34
A.20 osol.gpu_model 34
A.21 osol.action 35
A.22 osol.action_type 35
A.23 osol.document 36
A.24 osol.document_type 36
A.25 osol.sticker_print 36

B OSOL-T architecture - why use queue? 37

List of Tables

1 Sticker data - stage 0 18
2 Sticker data - stage 1 18
3 Sticker data - stage 1 - equip_info * - monitor 19
4 Sticker data - stage 1 - equip_info * - keyboard 19
5 Sticker data - stage 1 - equip_info * - mouse 19
6 Sticker data - stage 1 - equip_info * - laptop or tablet 20
7 Sticker data - stage 1 - cpu_info * 20
8 Sticker data - stage 1 - gpu_info * 20
9 Sticker data - stage 2 20
10 Sticker data - stage 3 21
11 conf.person_type content 27
12 conf.communication_mean content 29
13 conf.equip_type content 29
14 conf.action_type content 35

List of Figures

1 OSOL sticker sample 15
2 OSOL equipments lifecycle 17
3 OSOL Tracer system draft architecture[6] 21
4 OSOL database schema 22
5 MS Azure Serverless Architecture[7] 24
6 OSOL database metadata - table_audit_columns 26
7 OSOL database metadata - osol.person 26
8 OSOL database metadata - conf.person_type 26
9 OSOL database metadata - conf.language 27
10 OSOL database metadata - osol.address 27
11 OSOL database metadata - conf.country 28
12 OSOL database metadata - osol.communication 28
13 OSOL database metadata - conf.communication_mean 28
14 OSOL database metadata - osol.equip 29
15 OSOL database metadata - conf.equip_type 29
16 OSOL database metadata - osol.monitor 30
17 OSOL database metadata - osol.monitor_model 30
18 OSOL database metadata - osol.keyboard 31
19 OSOL database metadata - osol.keyboard_model 31
20 OSOL database metadata - osol.mouse 32
21 OSOL database metadata - osol.mouse_model 32
22 OSOL database metadata - osol.laptop 32
23 OSOL database metadata - osol.laptop_model 33
24 OSOL database metadata - osol.cpu_model 34
25 OSOL database metadata - osol.gpu_model 34
26 OSOL database metadata - osol.action 35
27 OSOL database metadata - osol.action_type 35
28 OSOL database metadata - osol.document 36

29 OSOL database metadata - osol.document_type 36
30 OSOL database metadata - osol.sticker_print 36

1 What is One Student One Laptop(OSOL)?

One Student One Laptop (OSOL) refurbishes donated, out-of-date IT equipment, including laptops, tablets, flat monitors, mouses, and keyboards. After cleaning and disinfecting the devices, we perform a factory reset. Where possible, we install ChromeOS Flex (a lightweight, cloud-based operating system) and a selection of educational content. These refurbished devices are then loaned free of charge to students in developing countries to support their education and personal growth.

OSOL stands out from other IT reuse organizations by focusing on education, sustainability, and accessibility for students. Here's how:

- **Education First:** OSOL is dedicated to helping students by providing technology specifically for learning.
- **Sustainable Tech Use:** Instead of just donating devices, OSOL recalls and replaces them annually, preventing e-waste and keeping technology up to date.
- **Free Lending Model:** Unlike many organizations that sell or give away refurbished devices, OSOL lends them for free, making access more equitable.
- **Online Learning Recommendation:** OSOL recomments students with trusted free online educational resources to enhance learning.
- **Offline Access:** For students without internet, OSOL pre-installs offline educational content and even AI-powered chatbot, ensuring they can still learn.

2 Why OSOL?

2.1 Education would change one life

Education is more than just learning facts and skills. It is a transformative process that can change your life and the lives of others. Education can help you discover your passions, develop your talents, and achieve your goals. Education can also empower you to make a positive difference in the world, by contributing to social change, economic growth, and human development.

But not everyone has access to quality education. Millions of people around the world face barriers such as poverty, discrimination, conflict, and lack of resources. These challenges prevent them from reaching their full potential and fulfilling their purpose.[4]

2.1.1 What is Education and Why is it Important?

Education is the process of acquiring knowledge, skills, values, and attitudes that enable people to learn, grow, and thrive. Education can take many forms, such as formal schooling, informal learning, vocational training, lifelong learning, etc. Education can also cover various subjects, such as literacy, numeracy, science, arts, culture, etc.

Education is important for many reasons. Here are some of them:

- **Education can improve your health and well-being.** Studies have shown that education can reduce the risk of diseases, increase life expectancy, improve mental health, and enhance self-esteem.

- **Education can increase your income and employment opportunities.** Studies have shown that education can boost your productivity, creativity, and innovation. Education can also help you access better jobs, earn higher wages, and escape poverty.

- **Education can enhance your social and civic participation.** Studies have shown that education can foster your critical thinking, communication, and collaboration skills. Education can also help you develop your values, beliefs, and identity. Education can also enable you to participate in democratic processes, such as voting, volunteering, and activism.

- **Education can promote peace and justice.** Studies have shown that education can reduce violence, conflict, and extremism. Education can also help you understand different perspectives, respect diversity, and promote human rights.

2.1.2 How Education Can Empower Individuals and Communities

Education is not only beneficial for individuals but also for communities. Education can empower people to take charge of their own lives and contribute to the common good. Here are some examples of how education can empower individuals and communities:

- **Education can empower children and youth.** Children and youth are the future of our society. They have the potential to shape the world for the better. But they also face many challenges, such as malnutrition, disease, child labor, child marriage, etc. By providing them with education, we can help them grow up healthy, happy, and safe. Education can also help them develop their cognitive, emotional, social, and physical abilities. Education can also enable them to express their opinions, aspirations, and creativity.

- **Education can empower marginalized groups.** Marginalized groups are those who are excluded, ignored or discriminated against because of their ethnicity, religion, disability, sexual orientation, etc. By providing them with education, we can help them overcome these challenges and achieve social inclusion. Education can also help them develop their identity, dignity, and respect. Education can also enable them to access their rights, opportunities, and resources.

- **Education can empower communities.** Communities are groups of people who share a common interest, location, or culture. By providing them with education, we can help them improve their quality of life, well-being, and resilience. Education can also help them develop their social capital, collective action, and problem-solving skills. Education can also enable them to participate in community development, environmental protection, and conflict resolution.

2.2 The Internet and Education

The internet has profoundly transformed education, revolutionizing how students learn and access knowledge. Its impact is undeniable, making it an indispensable tool for modern learning.[3]

The internet provides unparalleled access to a wealth of information, available anytime, anywhere. Constantly updated content ensures students have access to the most current information, often surpassing the limitations of traditional textbooks. This empowers students

to become independent learners, actively engaging with information and taking ownership of their education. Research supports this, demonstrating a correlation between internet access and improved academic performance.[10]

However, the digital divide presents a significant challenge. While many students have smartphone access, access to laptops and computers can be limited, particularly in underprivileged households, creating disparities in learning opportunities. Furthermore, the prevalence of social media among students underscores the need for digital literacy education and responsible internet use.[8]

2.2.1 Unprecedented Access to Information

Before the internet, accessing information required significant effort. Libraries and encyclopedias were the primary sources, often limiting access and requiring time-consuming research. The internet has revolutionized this, providing instant access to a vast repository of information, including scholarly articles, research papers, and multimedia resources. This accessibility not only saves time and effort but also fosters independent research and critical thinking skills.[1]

2.2.2 Enhanced Communication and Collaboration

The internet facilitates seamless communication and collaboration among students, teachers, and educators worldwide. Online platforms enable real-time interaction, enabling students to connect with peers, engage in discussions, and collaborate on projects regardless of geographical location. This fosters a global learning environment, exposing students to diverse perspectives and cultural experiences.[1]

2.2.3 Expanded Access to Educational Resources

The internet provides access to a wealth of educational resources, including online libraries, interactive learning platforms, and video lectures from renowned experts. These resources enrich the learning experience, offering alternative perspectives, hands-on simulations, and opportunities for in-depth exploration. Students can delve into topics of interest, develop critical thinking skills, and stay abreast of the latest advancements in their fields.

2.2.4 Online Learning and Certification

The internet has democratized education by providing access to a wide range of online courses and certification programs offered by prestigious institutions worldwide. These platforms offer flexibility and self-paced learning options, making education more accessible to students with diverse schedules and commitments. Interactive elements, such as quizzes, peer reviews, and discussion forums, enhance the learning experience and foster a sense of community among learners.[5]

2.2.5 Flexible Learning Environments

The internet empowers students to learn anytime, anywhere. No longer confined to traditional classrooms, students can access coursework from the comfort of their homes, libraries, or even while traveling. This flexibility accommodates diverse learning styles and enables students to create personalized learning environments that suit their individual needs and preferences.[5]

2.2.6 Beyond the Classroom

The internet extends the learning experience beyond the classroom. Virtual field trips to museums, historical sites, and scientific laboratories provide immersive experiences, fostering curiosity and broadening horizons. These digital explorations cultivate a sense of global citizenship and empathy, preparing students to thrive in an interconnected world.

2.2.7 Cost-Effective Learning Opportunities

The internet has revolutionized access to education by making it more affordable and accessible to a wider audience. Traditional higher education can be expensive, often leaving students with significant debt. The internet offers a compelling alternative, providing high-quality learning opportunities at a fraction of the cost.

For individuals seeking to acquire specific skills or knowledge, the internet is an invaluable resource. Numerous online platforms offer courses on a vast array of subjects, taught by qualified professionals from around the globe. These courses are significantly more affordable than traditional university programs, making education accessible to a broader demographic.

Here are some popular online learning platforms: Coursera, Khan Academy, Udemy, YouTube, EdX, Skillshare. These platforms cater to diverse learning styles and offer a variety of learning materials, including video lectures, interactive exercises, and quizzes. Many even provide certification upon course completion, enhancing the value proposition for learners.[9]

2.3 The Impact of Artificial Intelligence on Education

Artificial Intelligence (AI) have the potential to revolutionize education in several ways: [2]

2.3.1 Personalized Learning Experiences

GPT-powered systems can analyze student data, such as learning styles, strengths, and weaknesses, to provide personalized learning experiences. This allows for tailored instruction that adapts to individual needs, optimizing learning outcomes and increasing student engagement.

2.3.2 24/7 AI-Powered Assistance

AI-powered chatbots can provide students with on-demand support, answering questions, offering guidance, and providing access to learning materials. This 24/7 availability reduces the burden on teachers and ensures students have timely assistance when needed.

2.3.3 Enhanced Learning and Reduced Errors

GPT can assist students in various ways, from identifying and correcting errors in their work to providing constructive feedback. This can help students improve their writing, critical thinking, and problem-solving skills.

2.3.4 Streamlining Tasks and Fostering Independent Learning

GPT can help students streamline their tasks, such as summarizing research articles, generating ideas for creative projects, and translating information. This allows students to focus on deeper learning and develop independent learning skills.

3 Related Resources

3.1 One Laptop Per Child



The One Laptop Per Child (OLPC) project was a non-profit initiative that operated from 2005 to 2014 with the ambitious goal of providing every child in the developing world with a rugged, low-cost, low-power laptop computer. OLPC, Inc continues to operate, but the design and creation of laptops is no longer part of its mission. [13]

Key Concepts:

- **Affordable and Durable:** The core idea was to create a durable and affordable laptop specifically designed for children in developing countries, often lacking access to electricity and reliable internet.
- **Educational Focus:** The project aimed to transform education by providing children with access to information, fostering creativity, and enabling collaborative learning.
- **Community-Driven:** OLPC emphasized community involvement, encouraging children to learn from each other and use technology to solve local problems.

The XO Laptop: A key aspect of the project was the development of the XO laptop, a unique device featuring:

- **Low-power consumption:** Designed to run on minimal energy, often utilizing solar power or hand cranks.
- **Durability:** Built to withstand harsh conditions and rough handling.
- **Educational software:** Pre-loaded with educational software and designed for intuitive use.
- **Connectivity:** Incorporated features like mesh networking to allow laptops to connect and share data even without internet access.

Impact and Challenges: While the OLPC project had a significant impact in some regions, it also faced challenges:

- **High costs:** Achieving the initial goal of a $100 laptop proved difficult.
- **Sustainability:** Ensuring long-term maintenance and support for the laptops and their software in developing countries presented challenges.
- **Educational impact:** The actual impact on student learning outcomes varied significantly across different regions and implementations.

Legacy: Despite its challenges, the OLPC project has had a lasting impact, inspiring other initiatives focused on providing affordable technology to children in need. It also spurred innovation in low-cost computing and educational technology.

3.2 Labdoo Deutschland

Labdoo.org is a non-profit collaborative platform that allows unused laptops loaded with educational applications to reach schools throughout the world without incurring any economic cost and without generating additional CO_2 emissions to the Planet. Thanks to the collaboration of thousands of people around the globe, the Labdoo humanitarian platform has grown to support over 2,600 schools in more than 148 countries, deploying more than 300 operational hubs spread over the five continents and benefiting more than 825,000 students from around the globe.

Everyone involved in the Labdoo project is a volunteer and none of Labdoo members receives any remuneration for the time contributed to the cause, other than the satisfaction of seeing the joy of a child receiving education.

3.3 World Computer Exchange

World Computer Exchange (WCE) is a non-profit organization founded in 1999 by Timothy Anderson, headquartered in Hull, Massachusetts. Its mission is to reduce the digital divide for youth in developing countries by providing refurbished computers and educational resources, thereby enhancing communities and promoting environmentally responsible reuse of electronic equipment.

WCE operates through multiple chapters across the United States and Canada, collecting donated computers, refurbishing them, and preparing them for shipment to schools, libraries, orphanages, and community organizations in developing nations. Volunteers inspect and repair each computer, installing the Ubuntu operating system along with educational materials to facilitate learning, even without internet connectivity.

The organization has established partnerships with over 900 entities in 81 developing countries, delivering more than 30,000 computers and setting up approximately 2,675 computer labs by 2011. These efforts have connected numerous youth to the internet, providing access to educational content and opportunities.

WCE also offers programs like eCorps, which recruits volunteers to install computers and provide technical support at partner sites lacking local expertise. Additionally, the Computers for Girls (C4G) initiative focuses on providing technological training and STEM education to girls in countries such as Ghana, Liberia, Mali, Zambia, and Pakistan.

By refurbishing and distributing computers, WCE not only promotes education and technological skills among youth in developing countries but also contributes to environmental sustainability by preventing electronic waste from ending up in landfills.

According to UNESCO, it is North America's largest non-profit supplier of tested used computers to schools and community organizations in developing countries. [14]

3.4 Computer Aid International

Computer Aid International is a UK-based charity and social enterprise that works to bridge the digital divide by providing refurbished computers and other ICT equipment to educational institutions and non-

profit organizations in developing countries. Their headquarters is in London and offices in South Africa, their team works across projects spanning Africa, Latin America, Europe, and Australia.

Here are some key aspects of their work:

- **Providing refurbished computers:** They collect used computers from companies, data-wipe them, refurbish them, and then distribute them to organizations in need.
- **Focus on developing countries:** They primarily work with organizations in developing countries to improve access to technology and education.
- **Promoting reuse and circular economy:** They emphasize the importance of reusing technology to reduce e-waste and its environmental impact.
- **Supporting open source and internet neutrality:** They advocate for open-source software and net neutrality to ensure equitable access to information and technology.

3.5 Computer for School in Kenya

Computer For Schools Kenya (CFSK) is a charitable non-governmental organization registered in the year 2002, and now has eight Regional Centres hosted by various Government institutions country-wide. Their head office is located in Mihango Utawala, Embakasi, off the Eastern Bypass near the AP/GSU Training College Nairobi, Kenya.

In the 18 years of existence, CFSK has sourced for over 400,200 Personal Computers fully installed with requisite software and deployed in well over 12,010 Public Secondary and Primary schools, Technical Institutes, Teacher Training Colleges, Medical Training Centres and several Universities. They also operate a comprehensive technical maintenance support programme for these computers to ensure they are fully operational. In a number of these institutions, They have also placed power generators, established Local Area networks and Internet connectivity.

They have successfully developed digital multimedia teaching and learning resources specifically intended for our national Secondary School curriculum – providing both teachers and students with an invaluable modern tool that makes learning stimulating. They have also carried out computer proficiency training to over 100,000 school principals, Education Officers and teachers.

In a quest to preserve and conserve the environment, CFSK established the first e-waste management facility in Kenya – the WEEE Centre, for recycling and disposal of all electronic wastes.

Their organizational and operational model has been recognized as an exemplar, with ongoing efforts to replicate the same in other African countries such as Uganda, Tanzania, Burundi, Rwanda, Nigeria, South Sudan, Malawi and Namibia.

CFSK is a partnership of communities, private sector corporations, civil society organizations, and international charities and development partners working together for the long term good of the nation.

CFSK's considerable success is directly attributed to these close partnerships, as a result of which an estimated 1.5 Million young Kenyans now have access to information and communication technologies. Essentially, CFSK plays a crucial role in empowering Kenyan communities by providing access to technology and promoting digital literacy, which are essential for social and economic development in the 21st century.

3.6 Camara Education

Camara Education is a non-profit social enterprise that provides technology and training to improve education in underprivileged communities. Founded in 2005, Camara refurbishes used computers, installs educational software, and delivers them to schools in Africa and Ireland. The organization also offers teacher training programs to enhance digital literacy and integrate technology into classrooms effectively.

Key Activities of Camara Education:

- **Refurbishing Computers:** Collects and refurbishes donated IT equipment, making it suitable for educational use.
- **Providing Digital Learning Solutions:** Supplies computers with pre-installed educational content tailored to local curricula.
- **Teacher Training:** Trains educators to use technology in teaching, improving digital skills and classroom engagement.
- **Bridging the Digital Divide:** Focuses on providing technology access to underserved communities, especially in Africa and disadvantaged areas in Ireland.
- **Sustainable & Ethical Practices:** Ensures responsible recycling of IT equipment to minimize environmental impact.

Since its inception, Camara Education has impacted millions of students and thousands of teachers, promoting digital inclusion and improving educational opportunities.

3.7 Close the Gap

Close the Gap is an international social enterprise that aims to bridge the digital divide by offering high-quality, pre-owned computers donated by European and international companies to educational, medical, and social projects in developing and emerging countries.

Close the Gap collects decommissioned computers from companies and arranges for other organizations to clean the hard disks and then check and configure the hardware according to the requirements of its end-users. The computers are then shipped to the destination country by sea or air transport, to support educational, medical, entrepreneurial, and social projects. All the projects are demand-driven and impact-oriented initiatives

Since 2003, Close the Gap has already received more than 1 499 716 ICT devices from companies all over the world. Today, Close the Gap has supported more than 10 670 projects in more than 50 countries (mostly in Africa), and reached more than 5 496 807 beneficiaries.

Close the Gap values the sustainable recycling of e-waste. Through Worldloop, they make sure that the donated computers are disposed of in an environmentally responsible way when they are no longer useful. This platform ultimately aims at building a bridge between the Belgian private sector and the development sector in realizing the Sustainable Development Goals and creating new partnerships.

All of these initiatives contribute to creating impact in Africa and other developing countries.

3.8 Open Educational Resources



Open Educational Resources (OER) are freely accessible, openly licensed educational materials that can be used, adapted, and shared for teaching, learning, and research. OER includes textbooks, lecture notes, assignments, quizzes, videos, and other digital content designed to support education at all levels.

Traditional commercial educational materials, such as textbooks, are protected under conventional copyright terms. However, alternative and more flexible licensing options have become available as a result of the work of Creative Commons, a non-profit organization that provides ready-made licensing agreements that are less restrictive than the "all rights reserved" terms of standard international copyright. These new options have become a "critical infrastructure service for the OER movement." Another license, typically used by developers of OER software, is the GNU General Public License from the free and open-source software (FOSS) community. Open licensing allows uses of the materials that would not be easily permitted under copyright alone.Massive Open Online Courses (MOOCs) are free online courses available to anyone who wants to enroll.

Key Characteristics of OER

- **Free Access:** OER materials are available at no cost, removing financial barriers to education.
- **Open Licensing:** Most OER are released under licenses like Creative Commons, allowing users to modify, distribute, and use the content without restriction.
- **Customization & Adaptation:** Educators can adapt OER to fit specific teaching needs, updating content to stay relevant.

- **Digital & Print Formats:** OER exists in various formats, including PDFs, websites, videos, and interactive simulations.

Benefits of OER

- **Reduces Education Costs:** Provides free learning resources, reducing the need for expensive textbooks.
- **Enhances Teaching & Learning:** Encourages collaboration, innovation, and sharing among educators.
- **Expands Access to Quality Education:** Helps students and teachers worldwide, especially in underserved areas.
- **Supports Lifelong Learning:** Available to anyone, making education accessible beyond traditional classrooms.

Examples of OER Platforms

- **Khan Academy** (www.khanacademy.org) – Free courses on various subjects.
- **MIT OpenCourseWare** (ocw.mit.edu) – University-level courses and materials.
- **OpenStax** (openstax.org) – Free digital textbooks for higher education.
- **OER Commons** (www.oercommons.org) – A vast library of open educational resources.

3.9 eGranary Digital Library

The eGranary Digital Library is a program developed by the WiderNet project at the University of North Carolina at Chapel Hill. It's essentially a "library in a box" designed to provide access to educational resources in areas with limited or no internet connectivity.

Here's a breakdown of what it is and how it works:

- **Offline access to information:** The eGranary is a collection of digitized educational resources, including websites, books, videos, journals, and more, that can be accessed offline. It's pre-loaded onto a server (often a laptop or small computer) that creates a local network. Users in the community can then connect to this local network via Wi-Fi or Ethernet to access the vast library of information.
- **Curated content:** The content within the eGranary is carefully curated and selected to be relevant to the needs of the communities it serves. This often includes educational materials aligned with local curricula, health information, agricultural resources, and more.
- **Updates and expansion:** While the eGranary is designed for offline use, it can be updated periodically. New content can be added, and existing content can be refreshed when internet access becomes available.

- **Target audience:** The eGranary is particularly beneficial for schools, libraries, clinics, and other organizations in remote or underserved areas where internet connectivity is unreliable or non-existent. It empowers these communities by providing access to information that would otherwise be unavailable.

- **Open source and collaborative:** The eGranary project is open source, and WiderNet encourages collaboration and contributions from others. This allows for the customization and adaptation of the eGranary to meet the specific needs of different communities.

In short, the eGranary Digital Library is a powerful tool for bridging the information gap and promoting education and development in areas with limited internet access. It empowers communities by providing access to a wealth of knowledge and resources, even when they are offline.

3.10 UNESCO

Education is fundamental to UNESCO's mission to build a peaceful, equitable, and sustainable world. It is a fundamental human right for all, throughout life. As the UN agency with a mandate covering all aspects of education, UNESCO leads the Global Education 2030 Agenda, striving to achieve Sustainable Development Goal 4. [11]

3.10.1 The Importance of Digital Innovation in Education

In today's interconnected world, digital technology is crucial for ensuring equitable access to education. The COVID-19 pandemic starkly highlighted the digital divide, with many students lacking access to reliable internet and quality digital learning resources. This disruption underscored the urgent need to leverage technology to create more inclusive, resilient, and equitable education systems.

UNESCO supports the integration of digital innovation to:

- **Expand access to education:** Bridge the digital divide and ensure equitable access to quality education for all.

- **Enhance learning:** Improve the quality and relevance of learning by leveraging innovative technologies.

- **Foster lifelong learning:** Develop ICT-enhanced pathways for lifelong learning and skill development.

- **Strengthen education systems:** Build robust and resilient education systems that effectively utilize technology.

- **Develop digital literacy:** Equip teachers and students with the necessary digital skills and competencies.

3.10.2 The Role of Artificial Intelligence in Education

Artificial Intelligence (AI) presents both significant opportunities and challenges for education. While AI has the potential to personalize learning, automate administrative tasks, and improve educational outcomes, it is crucial to ensure that AI applications are inclusive, equitable, and human-centered.

UNESCO supports Member States in harnessing the potential of AI to achieve the Education 2030 Agenda. Key areas of focus include:

- **Addressing inequalities:** Ensuring AI does not exacerbate existing inequalities in access to education and opportunities.
- **Promoting inclusivity:** Ensuring AI technologies are accessible and inclusive for all learners, regardless of their background or abilities.
- **Guiding ethical AI development:** Promoting the ethical and responsible development and use of AI in education.

UNESCO's guidance documents, such as "Artificial Intelligence and Education: Guidance for Policy-makers", provide valuable insights for educators, policymakers, and practitioners on how to effectively and ethically integrate AI into education systems. [12]

4 What OSOL will benfit to Riverty and Bertelsmann?

Each loaned IT devices features several OSOL stickers prominently displaying the Riverty and Bertelsmann logos, as illustrated in Figure 1.



Figure 1. OSOL sticker sample

The OSOL sticker provides several advertising benefits:

- **Brand Visibility:**

  - **Increased Brand Awareness:** Every device with the sticker acts as a moving advertisement. Wherever the device goes, the logos are visible, increasing brand awareness among the recipients (students) and anyone who comes into contact with the device.

  - **Brand Association with Education:** By associating their brands with an initiative that provides educational technology to underprivileged students, both Riverty and Bertelsmann can enhance their corporate social responsibility image. This can positively impact public perception and attract potential customers who value socially conscious companies.

- Community Impact:

  - **Positive Brand Perception:** The sticker visibly demonstrates the companies' commitment to supporting education and improving the lives of children in developing countries. This can generate positive goodwill within the communities where the devices are distributed.

- Long-Term Exposure:

  - **Sustained Brand Visibility:** The stickers will remain on the devices for an extended period, providing ongoing brand exposure.

- Minimal Cost:

  - **Cost-Effective Advertising:** The sticker placement represents a highly cost-effective advertising method compared to traditional advertising channels.

The OSOL sticker serves as a low-cost, yet impactful, advertising tool for Riverty and Bertelsmann. It enhances brand visibility, strengthens their corporate social responsibility image, and fosters positive community relations.

5 How OSOL?

5.1 One IT equipment lifecycle

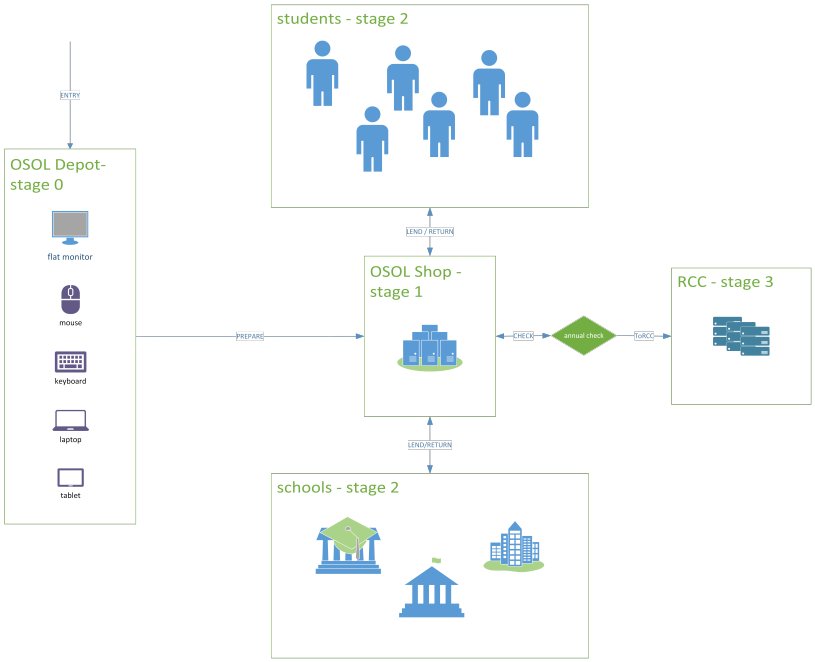


Figure 2. OSOL equipments lifecycle

5.1.1 Equipment stages

1. Stage 0 - In OSOL Depot
2. Stage 1 - In OSOL Shop
   - **Hardware:** basic repair, clean, disinfect
   - **Software:** factory reset, install ChromeOS Flex, install selected education content
3. Stage 2 - Distribution to Students
   - **less than Grade6 or 12 years old:** Tablet
   - **from Grade7 or 13 years old)** Laptop
4. Stage 3 - At Reiverty Computer Center (RCC)

5.1.2 Equipment score

- Each device have an integer score to express this device performance and usability.
- Each model (cpu, gpu, laptop, tablet, mouse, keyboard, monitor) have an integer score to express this IT model performance.
- The device score is decided by model score, production date and current equip status.
- Large score mentions high performance and good equip status.

5.1.3 Equipment annual check

- Each year we could decide one device score for each equip_type, the device which score is less than current-year-score will be disuseed.
- The disused deviced will be callbacked and send to Riverty Computer Center (RCC).

5.2 OSOL 2D barcode on IT equipment

5.2.1 Stage0 2D barcode data

1	stricker_print_id
2	equip_id
3	from_person_id
4	to_person_id
5	action_id
6	action_date

Table 1. Sticker data - stage 0

5.2.2 Stage1 2D barcode data

1	stricker_print_id
2	equip_id
3	from_person_id
4	to_person_id
5	action_id
6	action_date
7	equip_score
8	equip_info ∗

Table 2. Sticker data - stage 1

If equipment is moniter, equip_info ∗ is Table 3.
If equipment is keyboard, equip_info ∗ is Table 4.
If equipment is mouse, equip_info ∗ is Table 5.
If equipment is laptop or tablet, equip_info ∗ is Table 6.
The cpu_info ∗ is Table 7.
The gpu_info ∗ is Table 8.

1	brand
2	model
3	release_date
4	monitor_type
5	monitor_resolution
6	vga
7	dvi
8	dp
9	hdmi
10	usb_c
11	production_date

Table 3. Sticker data - stage 1 - equip_info $*$ - monitor

1	brand
2	model
3	release_date
4	is_wireless
5	is_bluetooth
6	is_usb_receiver
7	key_number
8	keyboard_type
9	production_date

Table 4. Sticker data - stage 1 - equip_info $*$ - keyboard

1	brand
2	model
3	release_date
4	is_wireless
5	is_bluetooth
6	is_usb_receiver
7	production_date

Table 5. Sticker data - stage 1 - equip_info $*$ - mouse

1	brand
2	model
3	release_date
4	cpu_info ∗
5	memory_type
6	memory_size(GB)
7	gpu_info ∗
8	disk_type
9	disk_size(GB)
10	monitor_type
11	monitor_size(inch)
12	monitor_resolution
13	is_wifi
14	is_bluetooth
15	usb_a_num
16	usb_c_num
17	hdmi_num
18	dp_num
19	vga_num
20	production_date

Table 6. Sticker data - stage 1 - equip_info ∗ - laptop or tablet

1	cpu_name
2	manufacturer
3	cores
4	clock(GHz)
5	L3_cache(MB)
6	release_date

Table 7. Sticker data - stage 1 - cpu_info ∗

1	gpu_name
2	manufacturer
3	gpu_chip
4	clock(GHz)
5	gpu_memory(MB)
6	fp32(TFLOPS)
7	release_date

Table 8. Sticker data - stage 1 - gpu_info ∗

5.2.3 Stage2 2D barcode data

1	stricker_print_id
2	equip_id
3	from_person_id
4	to_person_id
5	action_id
6	action_date

Table 9. Sticker data - stage 2

5.2.4 Stage3 2D barcode data

1	stricker_print_id
2	equip_id
3	from_person_id
4	to_person_id
5	action_id
6	action_date

Table 10. Sticker data - stage 3

5.3 OSOL preinstalled education content

5.3.1 Online education content

1. Math-desmos
2. OpenAI ChatGPT
3. Google Gemini
4. ResearchGate

5.3.2 Offline education content

5.4 OSOL Tracer - OSOL Data Management System

5.4.1 OSOL-T Architecture

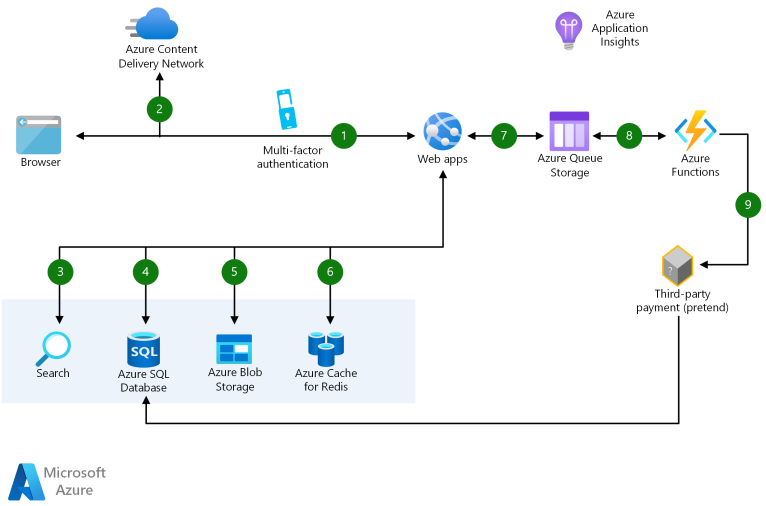


Figure 3. OSOL Tracer system draft architecture[6]

The OSOL-T system is an information management system with the following features:

- Records can be imported and accessed globally.
- Records are imported and retrieved individually via a graphical user interface (GUI).

- The OSOL-T GUI is accessible on any device (web, smart devices, etc.).
- Time performance is not a critical requirement for OSOL-T.

Given these requirements, a MS Azure Serverless architecture (as depicted in Figure 3) has been chosen for the OSOL-T system.

5.4.2 OSOL-T Database

- OSOL-T database is a relational database.
- OSOL-T database will be deployed in MS Azure SQL Server.



Figure 4. OSOL database schema

The OSOL-T database (Figure 4), detailed in Appendix A, comprises the following tables:

- **osol.person:** Stores records for all individuals associated with the system, including device owners, OSOL staff, students, and RCC staff. Address and communication information for each person are stored in the related tables:

  - **osol.address**
  - **osol.communication**

- **osol.equip:** Contains information on all devices within OSOL, including their current stage and owner. Device-specific details are stored in the following tables, categorized by equipment type:
  - **osol.laptop**
  - **osol.monitor**
  - **osol.mouse**
  - **osol.keyboard**

- **osol.action:** Records all activities related to a device, such as its entry into OSOL, preparation for lending, lending to a student, return to OSOL, and transfer to RCC. Associated activity types and possible documents are detailed in the following tables:
  - **conf.action_type**
  - **osol.document**

- **osol.sticker_print:** Stores information about each printed sticker. Stickers are printed in relation to a specific action and reflect the device's stage after that action. Multiple stickers can be printed for a single action.

5.4.3 OSOL-T API

- OSOL-T API is Restful API for each database object CRUD operations.
- OSOL-T API will be deployed as MS Azure Function

5.4.4 OSOL-T API Management

Azure API Management (APIM) is a powerful tool that helps organizations manage and secure their APIs. As showed in Figure 5, we could use APIM in OSOL-T system.

1. **Centralized API Management:**
   - **Organize and control:** APIM provides a single platform to manage all your APIs, regardless of where they're hosted (Azure, on-premises, other clouds).
   - **Simplify complexity:** It acts as a gateway, abstracting the complexities of your backend systems from the outside world.
   - **Consistent policies:** Apply consistent security, access control, and usage policies across all your APIs.
2. **Enhanced Security:**
   - **Protect your APIs:** APIM offers robust security features like authentication, authorization, rate limiting, and threat protection to safeguard your APIs from unauthorized access and attacks.
   - **Control access:** Implement fine-grained access control to restrict who can use your APIs and what they can do.

3. **Improved Developer Experience:**
    - **Easy discovery:** APIM provides a developer portal where developers can easily discover, explore, and learn about your APIs.
    - **Self-service:** Developers can access documentation, code samples, and testing tools to quickly integrate with your APIs.
    - **Streamlined onboarding:** Simplify the process for developers to start using your APIs.
4. **Analytics and Monitoring:**
    - **Gain insights:** APIM provides detailed analytics on API usage, performance, and errors, helping you understand how your APIs are being used and identify potential issues.
    - **Monitor performance:** Track API performance and availability to ensure your APIs are meeting your service level agreements.
5. **Monetization:**
    - **Create revenue streams:** If you want to monetize your APIs, APIM allows you to create subscription plans and track usage for billing purposes.

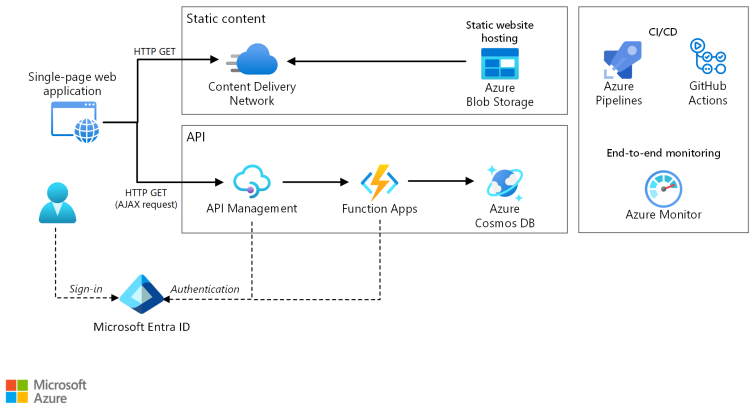


Figure 5. MS Azure Serverless Architecture[7]

5.4.5 OSOL-T GUI

OSOL-T GUI include features:

1. Multi-Platform - mobile, web - will be implemented with Flutter/Dart
2. Search
3. Equip page (CRUD)
    (a) Laptop page (CRUD)

        i. CPU page (CRUD)
        ii. GPU page (CRUD)
    (b) Monitor page (CRUD)
    (c) Keyboard page (CRUD)
    (d) Mouse page (CRUD)
4. Person page (CRUD)
    (a) Address page (CRUD)
    (b) Communication page (CRUD)
5. Action page (CRUD)
    (a) Print equip stage 2D barcode sticker from one action page
    (b) Document page (CRUD)
6. Scan 2D barcode, show data
7. AI-powered chatbot

5.5 Riverty Computer Center(RCC)

6 When OSOL - Schedule

6.1 Step 1

1. one place to store these out-of-date IT equipments.
2. two persons to monitor and confirm each other.

7 OSOL Budget

A OSOL-T Database metadata

A.1 table_audit_columns

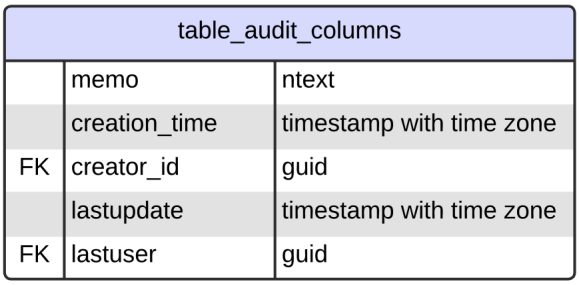


Figure 6. OSOL database metadata - table_audit_columns

A.2 osol.person

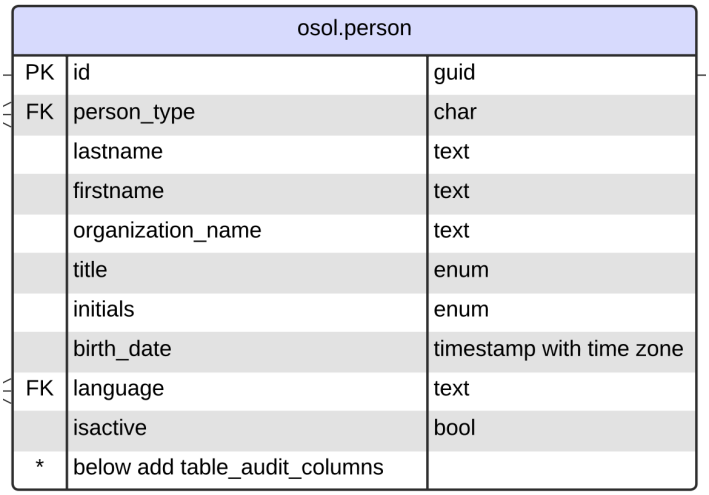


Figure 7. OSOL database metadata - osol.person

title enum: {Frau, Frau und Herr, Herr, Miss, Mr, Mrs, Ms }
initials enum: {Dr., Prof., Prof. Dr.}

A.3 conf.person_type

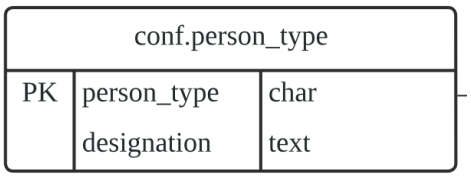


Figure 8. OSOL database metadata - conf.person_type

person_type	designation
C	company
F	female
M	male
U	unknown

Table 11. conf.person_type content

A.4 conf.language

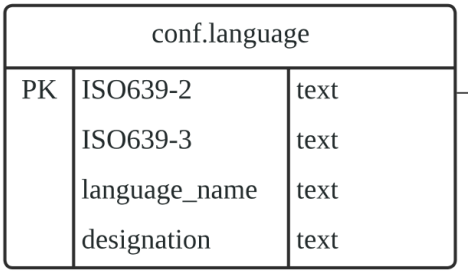


Figure 9. OSOL database metadata - conf.language

A.5 osol.address

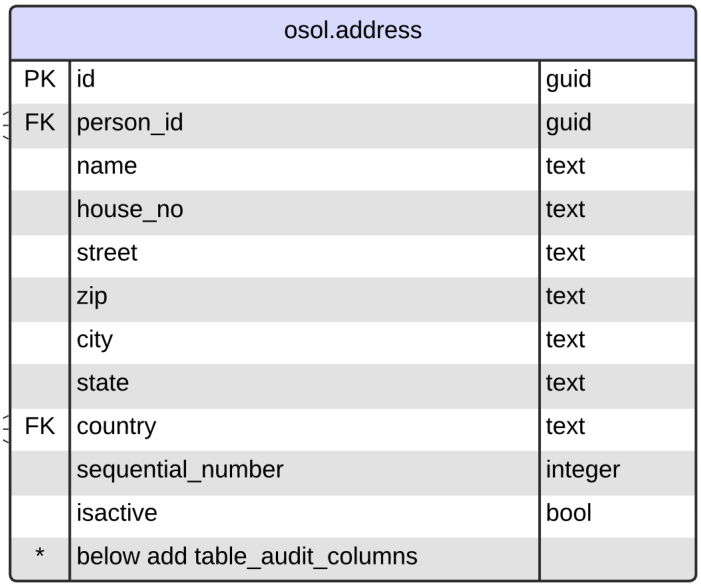


Figure 10. OSOL database metadata - osol.address

A.6 conf.country

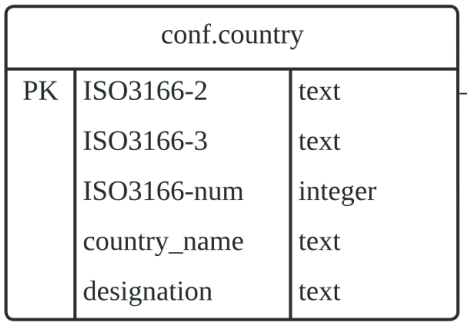


Figure 11. OSOL database metadata - conf.country

A.7 osol.communication

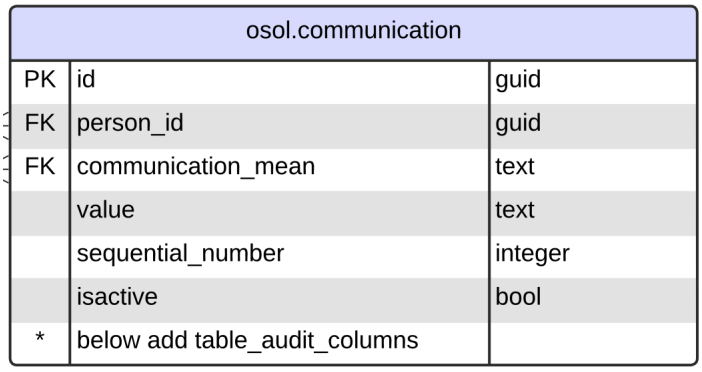


Figure 12. OSOL database metadata - osol.communication

A.8 osol.communication_mean

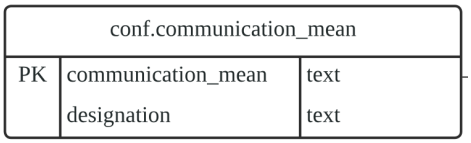


Figure 13. OSOL database metadata - conf.communication_mean

communication_mean	designation
telephone	T
mobilephone	T
fax	T
e-mail	E
website	

Table 12. conf.communication_mean content

A.9 osol.equip

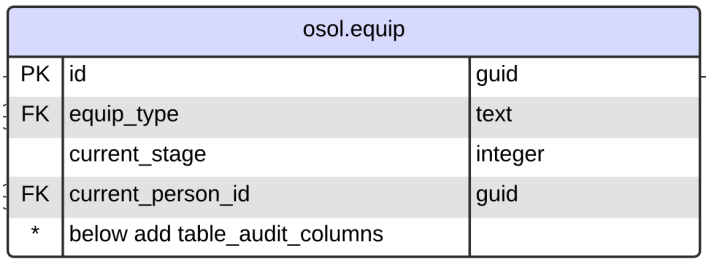


Figure 14. OSOL database metadata - osol.equip

A.10 conf.equip_type

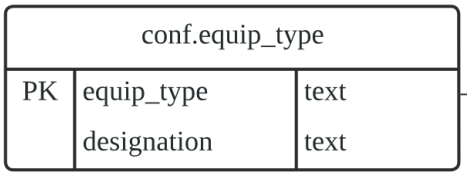


Figure 15. OSOL database metadata - conf.equip_type

equip_type	designation
laptop	desktop or laptop or tablet
mouse	mouse
keyboard	keyboard
monitor	monitor

Table 13. conf.equip_type content

A.11 osol.monitor

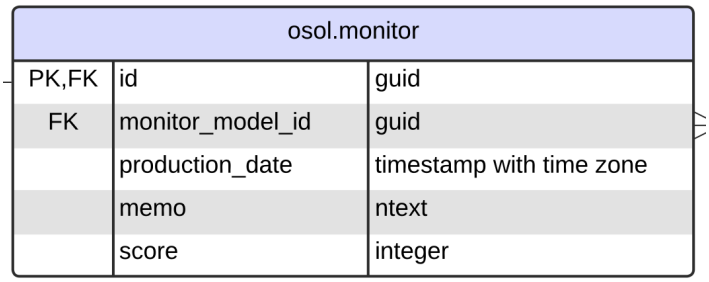


Figure 16. OSOL database metadata - osol.monitor

A.12 osol.monitor_model

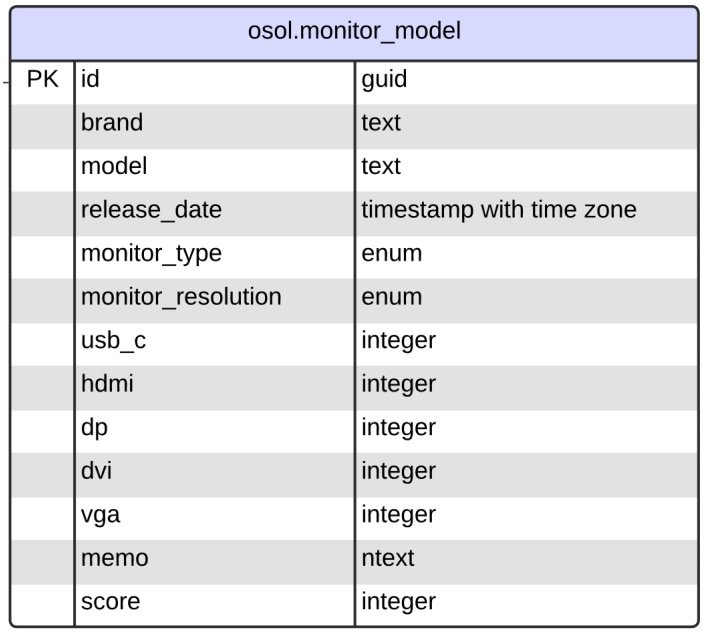
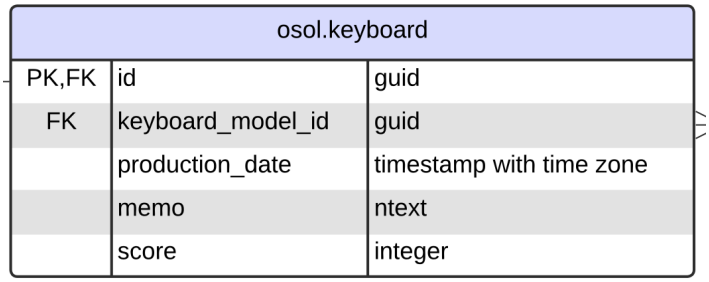


Figure 17. OSOL database metadata - osol.monitor_model

monitor_type enum: {TN, IPS, VA, LED, OLED, QLED}
monitor_resolution enum: {1024x768, 1280x720, 1280x800, 1280x1024, 1366x768, 1440x900, 1600x1200, 1920x1080, 2048x1536, 2560x1440, 3840x2160, 5120x2880, 7680x4320}

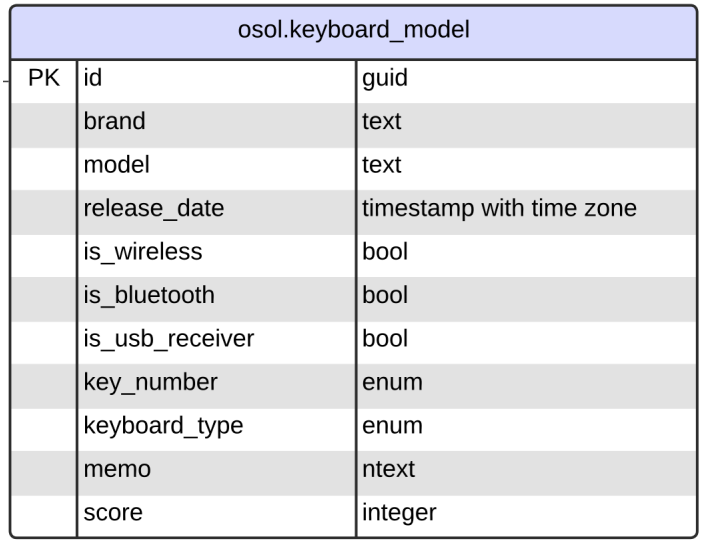
A.13 osol.keyboard



osol.keyboard		
PK,FK	id	guid
FK	keyboard_model_id	guid
	production_date	timestamp with time zone
	memo	ntext
	score	integer

Figure 18. OSOL database metadata - osol.keyboard

A.14 osol.keyboard_model



osol.keyboard_model		
PK	id	guid
	brand	text
	model	text
	release_date	timestamp with time zone
	is_wireless	bool
	is_bluetooth	bool
	is_usb_receiver	bool
	key_number	enum
	keyboard_type	enum
	memo	ntext
	score	integer

Figure 19. OSOL database metadata - osol.keyboard_model

key_number enum: {
Full-Size Keyboard (104–108 keys),
Tenkeyless (TKL) Keyboard (87–88 keys),
75% Keyboard (80–84 keys),
65% Keyboard (66–68 keys),
60% Keyboard (61 keys)
}
keyboard_type enum: {Mechanical, Membrane, Scissor-Switch, Optical & Hall Effect, Capacitive (Topre)}

A.15 osol.mouse

osol.mouse		
PK,FK	id	guid
FK	mouse_model_id	guid
	production_date	timestamp with time zone
	memo	ntext
	score	integer

Figure 20. OSOL database metadata - osol.mouse

A.16 osol.mouse_model

osol.mouse_model		
PK	id	guid
	brand	text
	model	text
	release_date	timestamp with time zone
	is_wireless	bool
	is_bluetooth	bool
	is_usb_receiver	bool
	memo	ntext
	score	integer

Figure 21. OSOL database metadata - osol.mouse_model

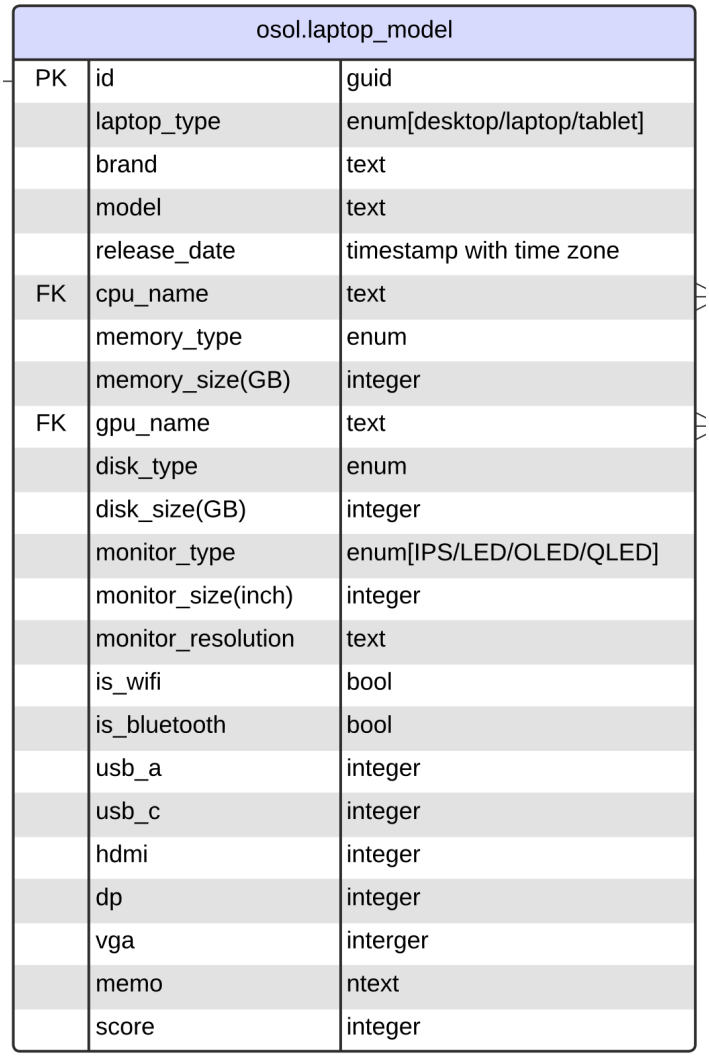
A.17 osol.laptop

osol.laptop		
PK,FK	id	guid
FK	laptop_model_id	guid
	os	enum
	production_date	timestamp with time zone
	memo	ntext
	score	integer

Figure 22. OSOL database metadata - osol.laptop

os enum: {ChromeOS Flex, Android, IPadOS}

A.18 osol.laptop_model



osol.laptop_model		
PK	id	guid
	laptop_type	enum[desktop/laptop/tablet]
	brand	text
	model	text
	release_date	timestamp with time zone
FK	cpu_name	text
	memory_type	enum
	memory_size(GB)	integer
FK	gpu_name	text
	disk_type	enum
	disk_size(GB)	integer
	monitor_type	enum[IPS/LED/OLED/QLED]
	monitor_size(inch)	integer
	monitor_resolution	text
	is_wifi	bool
	is_bluetooth	bool
	usb_a	integer
	usb_c	integer
	hdmi	integer
	dp	integer
	vga	interger
	memo	ntext
	score	integer

Figure 23. OSOL database metadata - osol.laptop_model

memory_type enum: {DDR3, DDR4, DDR5}
disk_type enum: {
SATA HDD,
SAS HDD,
IDE/PATA HDD,
SATA SSD,
NVMe SSD(PCIeSSD),
M.2 SSD,
U.2 SSD
}
monitor_type enum: {TN, IPS, VA, LED, OLED, QLED}

A.19 osol.cpu_model

osol.cpu_model		
PK	cpu_name	text
	manufacturer	enum[AMD/Intel/Apple/Other]
	cores	integer
	clock(GHz)	float
	L3_cache(MB)	integer
	release_date	timestamp with time zone
	memo	ntext
	score	integer

Figure 24. OSOL database metadata - osol.cpu_model

cpu_model manufacturer enum: {AMD, Intel, Apple, Other}

A.20 osol.gpu_model

osol.gpu_model		
PK	gpu_name	text
	manufacturer	enum[AMD/Intel/NVIDIA/ATI/Other]
	gpu_chip	text
	gpu_memory(GB)	integer
	clock(GHz)	float
	fp32(TFLOPS)	float
	release_date	timestamp with time zone
	memo	ntext
	score	integer

Figure 25. OSOL database metadata - osol.gpu_model

gpu_model manufacturer enum: {AMD, Intel, NVIDIA, ATI, Other}

A.21 osol.action

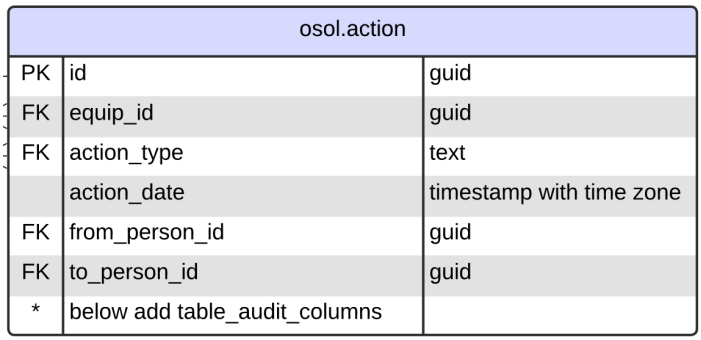


Figure 26. OSOL database metadata - osol.action

A.22 osol.action_type

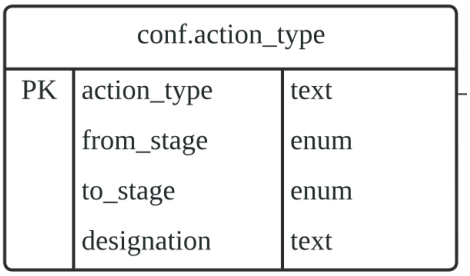


Figure 27. OSOL database metadata - osol.action_type

action_type	from_stage	to_stage	designation
entry	-1	0	
prepare	0	1	
lend	1	2	
return	2	1	
toRCC	2	3	
comm			record communicaiton, no stage changing
transfer			equip transfer from one owner to another owner, no stage changing

Table 14. conf.action_type content

A.23 osol.document

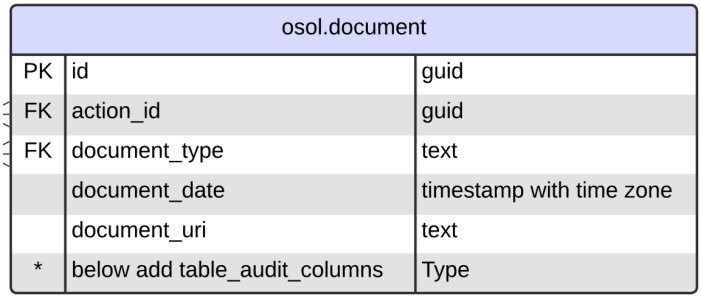


Figure 28. OSOL database metadata - osol.document

A.24 osol.document_type

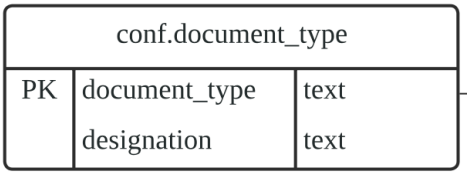


Figure 29. OSOL database metadata - osol.document_type

A.25 osol.sticker_print

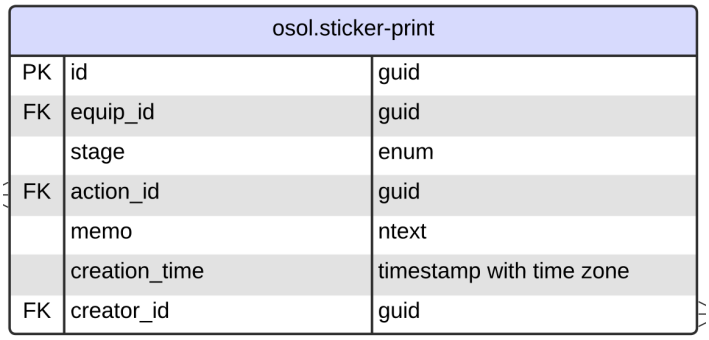


Figure 30. OSOL database metadata - osol.sticker_print

B OSOL-T architecture - why use queue?

1. Decoupling
    - Purpose: Queues act as a buffer between the API and the downstream services, such as databases or processing logic.
    - Benefit: This ensures that the API can handle incoming requests without being directly tied to the speed or availability of the downstream services.
    - Example: A user uploads data via an API, which places a message in a queue. The processing of the data happens asynchronously, allowing the API to quickly respond to the user.
2. Scalability
    - Purpose: In serverless environments, the workload can spike unpredictably. A queue helps handle bursts of traffic by storing messages until they can be processed.
    - Benefit: This allows your application to scale gracefully. For instance, Azure Functions or Logic Apps can process messages from the queue as they scale up or down automatically.
    - Example: During high traffic, the queue can hold requests until Azure Functions has enough capacity to process them.
3. Reliability and Resilience
    - Purpose: Queues ensure that no messages are lost, even if downstream services fail or experience downtime.
    - Benefit: This helps in building fault-tolerant systems where messages are retried or processed later when the system recovers.
    - Example: A queue can store order details submitted via an API. If the order-processing system crashes, the orders remain safe in the queue until the system is back online.
4. Throttling and Load Management
    - Purpose: Queues help control the rate at which messages are processed by downstream services, preventing overload.
    - Benefit: This protects the system from performance degradation or failures due to resource exhaustion.
    - Example: The API can accept thousands of requests per second, but the backend may process only a hundred at a time. The queue enforces this limit.
5. Asynchronous Processing
    - Purpose: Many tasks do not need to be performed immediately and can be handled in the background.
    - Benefit: This improves the user experience by providing faster responses from the API and ensures better resource utilization.

- Example: An image-processing API can accept uploads quickly and push them to a queue. Processing, resizing, and storing the images happens asynchronously.

6. Retry and Dead-Letter Handling
   - Purpose: Queues often support retries and dead-letter queues (DLQs) for failed messages.
   - Benefit: If a message cannot be processed, it can be retried a set number of times, and if it still fails, it is moved to a DLQ for further investigation.
   - Example: A payment processing API ensures no transaction request is lost. Failed requests are retried or stored in a DLQ.
7. Integration with Azure Services
   - Azure provides built-in support for queues with services like Azure Queue Storage and Azure Service Bus. These services integrate seamlessly with Azure serverless components such as:
     - Azure Functions: Triggered by queue messages to process tasks.
     - Azure Logic Apps: Orchestrates workflows using queue messages.
     - Azure Event Grid: Enables event-driven processing.
8. Real-World Scenario For an e-commerce API:
   (a) A user places an order via the API.
   (b) The order details are added to a queue.
   (c) Downstream services like inventory checking, payment processing, and order confirmation process the queue messages asynchronously.
   (d) If any service fails, the queue retains the message for retry or troubleshooting.

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